# LEAF Commons: Flexible Digital Tools and Responsive Scholarly Workflows (Workshop)

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This half-day workshop introduces textual scholars and practitioners to the LEAF Commons tool suite, a set of web-based, easy-to-use tools that support text encoding, named entity recognition, web annotation, text analysis and publication without users having to learn complex coding languages, and support the easy movement from one interoperable tool to the other depending on users' needs. This freely available suite of tools supports digital scholarly workflows for the collaborative production and publication of scholarly and documentary texts, editions, and collections on the web, without the need for software installation, promoting best practices for text encoding, annotation, and metadata standards. The LEAF Commons suite enables the use of individual tools for specific purposes, as well as supporting an end-to-end workflow beginning with outputs of optical character or handwriting recognition systems, transcriptions, or born-digital texts and ends with publication on the web, allowing it to serve a wide range of research and pedagogical uses. This workshop will introduce tools, features and functionality that have not before been available, integrate training materials in multiple languages, and encourage participants to think critically about how the LEAF Commons tools help researchers to make confident choices about their own workflows rather than committing to a full production and publication platform.

LEAF stands for the Linked Editing Academic Framework , a collaborative software suite that provides a set of modular tools for text editing and publication. The LEAF Commons tools constitute an accessible, low-barrier, no-cost infrastructure for researchers to produce online texts, editions, or collections, whether for teaching or for undertaking research and collaboration on a sustainable basis. The LEAF Commons tools are freely available in the browser, enabling collaboration and publication through Github, in addition to permitting local storage. Promoting the reuse of data in keeping with the FAIR (Findable, Accessible, Interoperable, and Reusable) data principles, LEAF uses open-source software, open-access platforms, and open international standards for best practices in text-encoding ( TEI-XML ) and web annotation ( RDF ).

LEAF promotes best practices, reuse, and sustainability in the production of digital scholarship in the humanities broadly by designing tools to bridge the gap between scholars who have coding and encoding experience, and those who do not. LEAF Commons offers communities of researchers, teachers, and students the opportunity to take part in digital knowledge production and open collaboration. The workshop will end with an open discussion about pursuing such forms of open knowledge production and collaboration.

**LEAF-Writer** : an open-source, open-access Extensible Markup Language (XML) editor that runs in a web browser and offers scholars and their students a rich textual editing experience without the need to download, install, and configure proprietary software, pay ongoing subscription fees, or learn complex coding languages. This user-friendly editing environment incorporates Text Encoding Initiatives (TEI) and Resource Description Framework (RDF) standards, meaning that texts edited in LEAF-Writer are interoperable with other texts produced by the scholarly editing community and with other materials produced for the Semantic Web. It also incorporates Named Entity Recognition and reconciliation with, or linking to, linked open data identifiers through the incorporation of the NERVE tool. LEAF-Writer is particularly valuable for pedagogical purposes, allowing instructors to teach students best practices for encoding texts without also having to teach students how to code. LEAF-Writer is designed to help bridge the gap by providing access to all who want to engage in new and important

forms of textual production, analysis, and discovery.

**DToC** : (the Dynamic Table of Contexts) provides an online interactive reading and publication environment for digital scholarly texts where the two conventional overviews provided in print editions - the table of contents and the index - have been dynamically merged to provide an interactive online e-reading experience that leverages the power of XML markup. Users can build a DToC edition from one or more TEI-XML files, then curate and label the underlying elements and attributes in order to understand where named entities, topics, and concepts can be traced within the edition. Editions can be stored using URLs and shared with readers as published or teaching texts.

**LEAF-TE** : (The LEAF Turning Engine) is a web interface that enables users to easily and automatically transform documents between formats. It converts HTR/OCR output and MS Word to TEI-XML for importing into LEAF-Writer. It also exports from LEAF-Writer to HTML, Markdown, RDF, CSV, and plain text for exporting encoded texts for publishing, analysis, and preparing for linking textual data.

NERVE **:** (the Named Entity Relationship and Vetting Environment) is an application that performs Named Entity Recognition (NER) on machine-readable texts, allowing users to identify candidate entities in a document, review, and correct the results . NERVE suggests relevant Uniform Resource Identifiers (URIs) for entities, so users can reconcile data to an authority such as Wikidata or the Virtual International Authority File to provide the basis for Linked Open Data (LOD) Web Annotations. Users can export their reconciled data in TEI-XML or HTML formats to an online repository or to their desktop. NERVE can be used from within LEAF-Writer, or as a stand-alone tool.

# Agenda:

The workshop will be managed as follows:

**Introductions** (15 minutes)

**Set up** (15 minutes)

Access to LEAF Commons tools via GitHub

**Part 1** (45 minutes)

Encoding and annotating text objects using TEI and RDF with LEAF-Writer

BREAK

**Part 2** (30 minutes)

Using NERVE to recognize, process, and associate named entities with external linked data authorities

**Part 3** (30 minutes)

Publishing texts, editions, anthologies or collections with DToC

**Part 4** (15 minutes)

Using LEAF-TE for importing and exporting different file types to expand

**Conclusion** (15 minutes)

Adapting LEAF Commons tools for custom workflows

# Requirements and Availabilities:

Participants will need a web-enabled laptop computer (not a tablet or a mobile device). No software installation is required. Sample texts in multiple languages will be provided, which participants will have continued access to after the workshop; participants are also welcome to bring their own documents in TEI-conformant XML.

# Statement Of Commitment To Diversity:

Attention to diversity will be given the the selection of sample texts and images to ensure that a range of genders, races, and nationalities are represented. Participants are also welcome to bring their own content. LEAF tools are designed to meet accessibility standards as well as to make tools more readily usable by a broader and more diverse user base than is typical in the digital humanities.

# Facilitators and Workshop History:

The LEAF team has conducted similar workshops at disciplinary conferences and in consultation with project teams over the past three years (see bibliography, below); in fact, we submitted a similar workshop proposal for DH2024 that was accepted. However, because of a misidentification in registration that the workshop would be given in a hybrid fashion (although we had explained to the conference organizers that a hybrid workshop would not be possible), most who registered could not participate and only seven people could take part. Therefore, we hope that we will be able to reach out to conference attendees who were unable to participate in last year's workshop.

**Susan Brown** (University of Guelph - ) Professor, Canada Research Chair in Collaborative Digital Scholarship. Her work explores intersectional

feminism, literary history, semantic technologies, and online knowledge infrastructures. She directs the Orlando Project, the Canadian Writing Research Collaboratory, and the Linked Infrastructure for Networked Cultural Scholarship.

**James Cummings** (Newcastle University - ) is the Reader in Digital Textual Studies and Late-Medieval Literature at the School of English Literature, Language, and Linguistics at Newcastle University. His interests include the use of digital technology for scholarly editing and also late-medieval performance. He has a long history of involvement with the TEI Consortium and is currently an elected member to the TEI's Board of Directors. He has recently been leading projects on both HTR to TEI workflows and digital pedagogy, as well as a sub-project at Newcastle using LEAF.

**Mihaela Ilovan** (Canadian Writing Research Collaboratory - ) Assistant Director. A librarian and digital humanist by formation, she is the technical project manager for the development of the LEAF Virtual Research Environment and the coordinator for LEAF data ingestion and mapping.

**Diane Jakacki** (Bucknell University - ) Digital Scholarship Coordinator, associated faculty in Comparative & Digital Humanities. At Bucknell she works with faculty, students, and their collaborators within and beyond Bucknell to develop and implement an array of text-centric multicultural and multilingual research projects. She is lead investigator of the LAB Cooperative, REED London, and currently serves as the chair of ADHO's Executive Board and the TEI's Board of Directors.

**Rachel Milio** (University of Crete - ) Doctoral candidate who has worked as research associate on the LEAF project and REED London for the past four years.

# Bibliography


**Brown, Susan** (2014): "Scaling Up Collaboration Online: Towards a Collaboratory for Research on Canadian Writing," in *International Journal of Canadian Studies* 48: 233-51.

**Brown, Susan / Simpson, John / the INKE Research Group / the CWRC project team** (2015): "The Changing Culture of Humanities Scholarship: Iteration, Recursion, and Versions in Scholarly Collaboration Environments; Special Issue on "Building Partnerships to Transform Scholarly Publishing," in *Scholarly and Research Communication* 5.4. < https://src-online.ca/index.php/src/article/view/191 >

**Brown, Susan / Cummings, James / Frizzera, Luciano / Ilovan, Mihaela / Jakacki, Diane / Milio, Rachel** (2024): "LEAF Commons: Using Linked Tools for Digital Editorial Production," workshop at the DH2024 Annual Meeting.

**Brown, Susan / Cummings, James / Frizzera, Luciano / Ilovan, Mihaela / Jakacki, Diane / Milio, Rachel** (2022): "Engaging TEI Editors Through LEAF-Writer," workshop at the TEI 2022 Annual Meeting.

**Brown, Susan / Jakacki, Diane** (2024): "'I don't even see the code:' The Dynamic Table of Contexts – Optimizing our Encoded Texts for Web-based Reading and Analysis Environments." TEI Annual Meeting.

—. "The Linked Editing Academic Framework (LEAF) in the multimodal annotation ecosystem." (2024) Reimagining Annotation for Multimodal Cultural Heritage. Rennes, France.

**Calarco, Gabriel / Cummings, James / Jakacki, Diane (2024)** "Producing Semantic Digital Editions Using LEAF Commons Tools" workshop at the TEI 2024 Annual Meeting. (Conducted in English and Spanish).

**Canning, Erin / Brown, Susan / Roger, Sarah / Martin, Kim** (2022): "The Power to Structure: Making Meaning from Metadata through Ontologies," in *Kula* 6.3.

**Cope, Bill, et al.**, eds (2011): *Towards a Semantic Web: Connecting Knowledge in Academic Research*. Chandos Publishing.

**Cummings, James** (2019): "Opening the Book: Data Models and Distractions in Digital Scholarly Editing." *International Journal of Digital Humanities*. 1, 179-193 (2019)

**Cummings, James / Flex, Valentina / Healey, Alexandra / Jakacki, Diane / Johnson, Ian / Pirmann, Carrie** (2022): "Evolving Hands: HTR and TEI Workflows for cultural institutions," at the TEI 2022 Annual Meeting.

**Cummings, James / Jakacki, Diane** (2023): "Building Workflows for HTR to TEI Up Conversion and Enhancement." at the TEI Annual Meeting.

**Jakacki, Diane (2022):** "Complex Textual Editing through Text Encoding: LEAF-Writer." Atelier "Éditions critiques numériques et multilinguisme," workshop at the Université de Montréal.

**Jakacki, Diane / Brown, Susan / Cummings, James / Ilovan, Mihaela** (2023): "LEAF: Developing Streamlined Digital Scholarly Workflows with the Linked Editing Academic Framework" workshop at the Digital Humanities 2023 Annual Meeting.

—. (2022): "The Linked Editing Academic Framework: Creating an Editorial Environment for Collaborative Scholarship and Publication," at the Digital Humanities 2022 Annual Meeting. < https://youtu.be/jyCXRrfh5vk>.

LEAF-Writer Commons: < https://leaf-writer.leaf-vre.org/>

LEAF-Writer Commons documentation: < https://portal.lincsproject.ca/docs/tools/leaf-writer/leaf-writer-documentation >

Linked Infrastructure for Networked Cultural Scholarship (LINCS) project. < https://lincsproject.ca >.

NERVE (Named Entity Recognition Vetting Environment. < https://portal.lincsproject.ca/docs/tools/nerve />.

Text Encoding Initiative (TEI). < https://tei-c.org/ >

The Resource Description Framework (RDF). < https://www.w3.org/RDF />